Kinematics

Questions 1–83

2.57 — Question 1. The points A and B have position vectors of $\begin{pmatrix}-2\\4\end{pmatrix}$ metres and $\begin{pmatrix}8\\11\end{pmatrix}$ metres respectively. Find the distance between A and B.

2.57 — Question 2. Given that a particle travels at the constant velocity given, find the speed of the particle:

a) $(-2\mathbf{i}+4\mathbf{j})$ ms^{-1}

b) $\begin{pmatrix}5\\8\end{pmatrix}$ ms^{-1}

c) $(2.5k\mathbf{i}-6k\mathbf{j})$ ms^{-1}, where $k>0$

2.57 — Question 3. A particle travels with a constant acceleration of $(-2\mathbf{i} + 3\mathbf{j})$ ms^{-2} from point A to point B. Its total displacement is $(-400\mathbf{i} + 250\mathbf{j})$ m. Its final velocity is $(12\mathbf{i} + 10\mathbf{j})$ ms^{-1}. Find

a) the magnitude of the acceleration of the particle.
b) the distance from A to B.
c) the speed of the particle when it passes through B.

2.57 — Question 4. A particle moves with constant acceleration $(4\mathbf{i} + 7\mathbf{j})$ ms^{-2}, and initially has velocity $(\mathbf{i} + 2\mathbf{j})$ ms^{-1}.
Find the velocity of the particle at $t = 3$ seconds.

2.57 — Question 5. A particle moves with constant acceleration $(-\mathbf{i} + 3\mathbf{j})$ ms^{-2}. It has velocity $(-2\mathbf{i} + 5\mathbf{j})$ ms^{-1} at $t = 7$ seconds. Find

a) the initial velocity of the particle.

b) the initial speed of the particle.

2.57 — Question 6. An object is moving with constant acceleration. Initially it has velocity $\begin{pmatrix} 4 \\ 3 \end{pmatrix}$ ms^{-1} and 7 seconds later it has velocity $\begin{pmatrix} 8 \\ 18 \end{pmatrix}$ ms^{-1}.
Find the magnitude of its acceleration.

2.57 — Question 7. Find the velocity vector of a particle in each case, given the speed v and direction in which the particle is travelling, leaving your answers in exact form:

a) $v = 10$ ms^{-1} parallel to $\begin{pmatrix}1\\0\end{pmatrix}$

b) $v = 4$ ms^{-1} parallel to $\begin{pmatrix}1\\1\end{pmatrix}$

c) $v = 0.3$ ms^{-1} parallel to $\begin{pmatrix}2\\-3\end{pmatrix}$

2.57 — Question 8. The position vector, $\mathbf{r}$ metres, of an object at time t seconds is given by

$$\mathbf{r} = t^2\mathbf{i} + (3 - 2t)\mathbf{j}$$

Find its displacement between $t = 2$ and $t = 5$

2.57 — Question 9. An object is initially at the origin moving with velocity $(2\mathbf{i} + 5\mathbf{j})$ ms^{-1}. It has a constant acceleration of $(6\mathbf{i} - \mathbf{j})$ ms^{-2}. Find

a) the position vector of the object at time t

b) the distance it is from the origin after 10 seconds.

2.57 — Question 10. A particle P moves with constant acceleration $(5\mathbf{i} + 4\mathbf{j})$ ms^{-2}, having been initially at rest at initial position $(2\mathbf{i} - 7\mathbf{j})$ metres.

a) Show that the expression for the position vector of P at time t seconds is given by

$$\mathbf{r} = \left(2 + \frac{5}{2}t^2\right)\mathbf{i} + (2t^2 - 7)\mathbf{j}$$

b) Hence, find the distance the particle is from the origin after 2 seconds, giving your answer to 3 significant figures.

2.57 — Question 11. Initially a particle position is $(6\mathbf{i} + 2\mathbf{j})$ m relative to a fixed origin O and is moving with velocity $(\mathbf{i} - 3\mathbf{j})$ ms^{-1} and constant acceleration $(8\mathbf{i} - 5\mathbf{j})$ ms^{-2}.
Find the position vector of the particle at $t = 4$ seconds.

2.57 — Question 12. A particle moves with constant acceleration $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ ms^{-2} and passes through the point A with velocity $\begin{pmatrix} -1 \\ 3 \end{pmatrix}$ ms^{-1}.
It then reaches point B 4 seconds after leaving A, where point B has position vector $\begin{pmatrix} 8 \\ -2 \end{pmatrix}$ metres relative to the origin.
Find the position vector of A.

2.57 — Question 13. A particle moves with constant acceleration $(2\mathbf{i} - 4\mathbf{j})$ ms^{-2}. At $t = 0$ it passes through the origin and T seconds later it passes through a point with position vector $(-12\mathbf{i}+k\mathbf{j})$ metres at velocity $(4\mathbf{i}+3\mathbf{j})$ ms^{-1}. Find

a) the value of T
b) the value of k

2.57 — Question 14. A particle P moves in a straight line from position $\begin{pmatrix}5\\-2\end{pmatrix}$ metres to $\begin{pmatrix}k\\78\end{pmatrix}$ metres, moving parallel to the vector $\begin{pmatrix}7\\10\end{pmatrix}$. Find the value of k

2.57 — Question 15. For each of the following, the two vectors represent the velocity of two particles which are moving with constant velocity and parallel to each other. In each case, find the value of k for this to hold:

a) $(2\mathbf{i} - 4\mathbf{j})$ ms^{-1} and $(4\mathbf{i} - k\mathbf{j})$ ms^{-1}

b) $(5\mathbf{i} + 3\mathbf{j})$ ms^{-1} and $(-10\mathbf{i} + k\mathbf{j})$ ms^{-1}

c) $(-3\mathbf{i} + 10\mathbf{j})$ ms^{-1} and $(k\mathbf{i} + 30\mathbf{j})$ ms^{-1}

d) $(1.5\mathbf{i} - 9\mathbf{j})$ ms^{-1} and $k(2\mathbf{i} - 12\mathbf{j})$ ms^{-1}

2.57 — Question 16. A particle passes through the origin at $t = 0$ with velocity $\begin{pmatrix}9\\2\end{pmatrix}$ ms^{-1} and constant acceleration $\begin{pmatrix}2\\3\end{pmatrix}$ ms^{-2}. At $t = T$ the particle is moving in the direction of $\begin{pmatrix}4\\1\end{pmatrix}$

Find the value of T

2.57 — Question 17. Emily is cycling with constant velocity $(4.2\mathbf{i} + 14.4\mathbf{j})$ ms^{-1} along a straight road. Henry is cycling parallel to Emily, on the same path.
Find the velocity for Henry in each of the following scenarios:

a) Henry is cycling twice as fast as Emily in the same direction.
b) Henry is cycling with speed 5 ms^{-1} in the same direction.
c) Henry is cycling with speed 12 ms^{-1} in the same direction.
d) Henry is cycling with speed 7.5 ms^{-1} in the opposite direction.

2.57 — Question 18. Two particles A and B are moving parallel to each other. Particle A is moving with constant velocity $\begin{pmatrix} 9 \\ 21.6 \end{pmatrix}$ ms^{-1}. Particle B begins at a position of $\begin{pmatrix} 2 \\ -3 \end{pmatrix}$ m relative to a fixed origin and moves with constant speed 19.5 ms^{-1}.
Find

a) the speed of Particle A.
b) the velocity of Particle B.
c) the position vector of B after 5 seconds.

2.57 — Question 19. Two particles A and B are moving parallel to each other, starting 4 metres apart and then at a fixed time later, are 4 metres apart again. Particle A travels a distance of 16 metres and particle B travels a distance of 20 metres.
Find the shortest distance between their two lines of motion, giving your answer to 3 significant figures.

2.57 — Question 20. Two squirrels are running along parallel branches of trees in the woods. At time $t = 0$ seconds, the first squirrel starts at position $(-\mathbf{i} + 3\mathbf{j})$ metres and is moving with constant speed of 5.2 ms^{-1}. The second squirrel runs with constant velocity $(2.4\mathbf{i} + 5.76\mathbf{j})$ ms^{-1}.
By modelling the branches as flat straight paths, find

a) the speed of the second squirrel.

b) the position vector of the first squirrel when 2.5 seconds have passed.

Given that the second squirrel starts at position $(11\mathbf{i} - 2\mathbf{j})$ metres,

c) determine the shortest distance between their two branches.

2.57 — Question 21. Particle A starts with position vector $\begin{pmatrix}-7\\-1\end{pmatrix}$ m and constant velocity $\begin{pmatrix}6\\3\end{pmatrix}$ ms^{-1}. Particle B starts with position vector $\begin{pmatrix}0\\2\end{pmatrix}$ m and constant velocity $\begin{pmatrix}2.5\\1.5\end{pmatrix}$ ms^{-1}.
Show that the particles collide.

2.57 — Question 22. Particles A and B are both moving in straight lines. Particle A has a constant velocity of $\begin{pmatrix}3.5\\12\end{pmatrix}$ ms^{-1} and Particle B moves with a speed of 6.25 ms^{-1}.
Both particle A and particle B pass through point P. Particle B reaches point P first, with particle A passing through it 2 seconds later. Given that the two lines of motion are perpendicular to each other, find the distance between particle A and particle B 6 seconds after particle B passed through point P.

2.57 — Question 23. The following questions that use i and j are unit vectors pointing east and north respectively. For each of the following velocity vectors, state the direction that the particle is moving in:

a) $(2\mathbf{i})$ ms^{-1}
b) $(-5\mathbf{j})$ ms^{-1}
c) $(4\mathbf{i} + 4\mathbf{j})$ ms^{-1}
d) $(-5\mathbf{i} + 5\mathbf{j})$ ms^{-1}

2.57 — Question 24. A ship, moving with constant acceleration, is travelling due east at a speed of 4 ms^{-1}. After 40 seconds it is travelling due south at 2 ms^{-1}. Find

a) the displacement from its initial position after 40 seconds.
b) the distance the ship has travelled from its initial position after 40 seconds.

2.57 — Question 25. The velocity, in ms^{-1}, of an object at time t seconds is given by

$$\mathbf{v} = (t-5)\mathbf{i} + (2t-13)\mathbf{j}$$

a) Show the object is never instantaneously at rest.
b) Find the time at which the object is moving directly to the south.
c) Find the time at which the object is moving to the north-east.

2.57 — Question 26. An object moves so that its position vector, in metres, relative to the origin O at time t seconds is given by

$$\mathbf{r} = (105 + 8t)\mathbf{i} + t^2\mathbf{j}$$

a) Determine the time at which the object is east of the origin.
b) Determine the time at which the object is north-east of the origin.
c) Explain why the object never moves in the westwards direction.

2.57 — Question 27. A particle passes point A with velocity $\begin{pmatrix}3\\8\end{pmatrix}$ ms^{-1} and has acceleration $\begin{pmatrix}2\\-20\end{pmatrix}$ ms^{-2}.
At the instant it reaches point B it is travelling in a due east direction.
Find the time at which it reaches point B

2.57 — Question 28. A particle passes point A with velocity $\begin{pmatrix}1\\3\end{pmatrix}$ ms^{-1} and has acceleration $\begin{pmatrix}15\\5\end{pmatrix}$ ms^{-2}.
At the instant it reaches point B it is travelling in a due north-east direction.
Find the time at which it reaches point B.

2.57 — Question 29. A particle is travelling with constant acceleration. It passes point A with velocity $\begin{pmatrix} 6 \\ -12 \end{pmatrix}$ ms^{-1} and then passes point B with velocity $\begin{pmatrix} -2 \\ 2 \end{pmatrix}$ ms^{-1}
Find

a) the acceleration of the particle when $t = 2$
b) the time at which it is travelling due south.
c) the speed at which it is travelling at the exact point it is travelling due south.

2.57 — Question 30. A particle is travelling with constant acceleration $(-\mathbf{i} - 3\mathbf{j})$ ms^{-2} and passes the origin with velocity $(13\mathbf{i} - 5\mathbf{j})$ ms^{-1}. After a further 5 seconds have passed, the particle reaches point P. Find

a) the speed the particle is travelling in when it passes through point P giving your answer to 3 significant figures.
b) Find the bearing at which the particle is travelling when it passes through point P

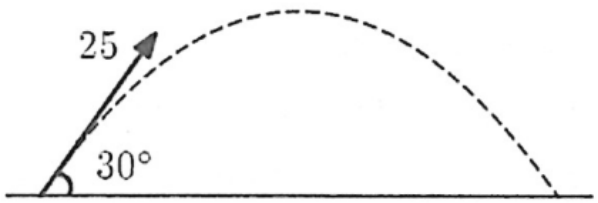
2.57 — Question 31. A helicopter is initially hovering above a field. It then sets off so that its acceleration is $(0.4\mathbf{i} + 0.2\mathbf{j})$ ms^{-2}.
The helicopter does not change its height as it moves.
Find

a) the speed of the helicopter 15 seconds after it leaves its position above the field.
b) the bearing on which the helicopter is travelling, 15 seconds after it leaves its position above the field, giving your answer correct to the nearest degree.

The helicopter stops accelerating when it is 240 metres from its initial position.

c) Find the time that it takes for the helicopter to travel from its initial position to the point where it stops accelerating.

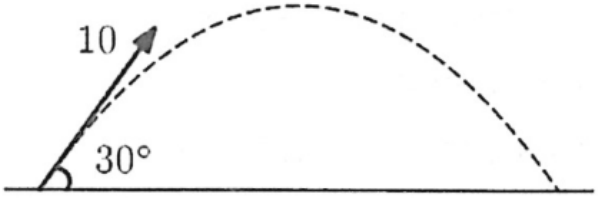
2.58 — Question 1. The following questions use $g = 9.8$ ms^{-2} A particle is projected from point O on horizontal ground with a velocity of 25 ms^{-1} at an angle of 30° above the horizontal before hitting the ground.



Find

a) the velocity of the particle after 0.5 seconds.
b) the greatest height the particle reaches.
c) the time the particle spends in the air.
d) the total horizontal distance travelled by the particle.

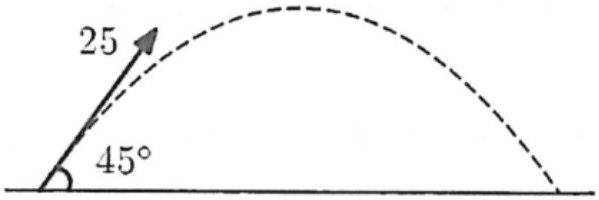
2.58 — Question 2. A particle is projected from point O on horizontal ground with a velocity of 10 ms^{-1} at an angle of 30° above the horizontal before hitting the ground.



Find

a) the velocity of the particle after 0.4 seconds.
b) the greatest height the particle reaches.
c) the time the particle spends in the air.
d) the total horizontal distance travelled by the particle.

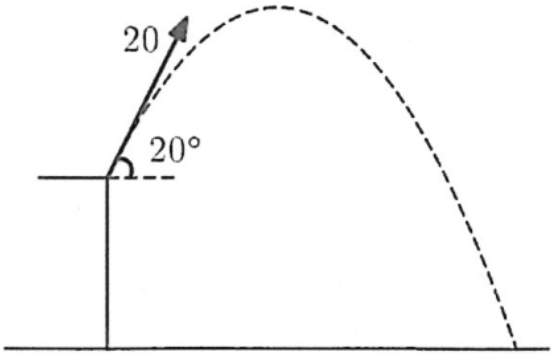
2.58 — Question 3. A particle is projected from point O on horizontal ground with a velocity of 25 ms^{-1} at an angle of 45° above the horizontal before hitting the ground.



Find

a) the velocity of the particle after 3 seconds.
b) the greatest height the particle reaches.
c) the time the particle spends in the air.
d) the total horizontal distance travelled by the particle.

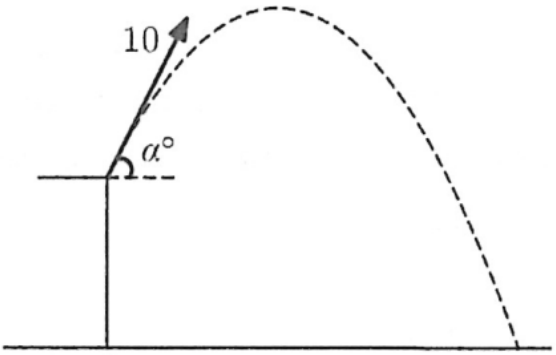
2.58 — Question 4. A ball is projected at an angle of 20° above the horizontal from a tower of 10 metres high at a speed of 20 ms^{-1} before landing on the floor.



Modelling the ball as a particle, find

a) the initial velocity of the ball.
b) the time the ball spends in the air.
c) the velocity of the ball as it hits the ground.

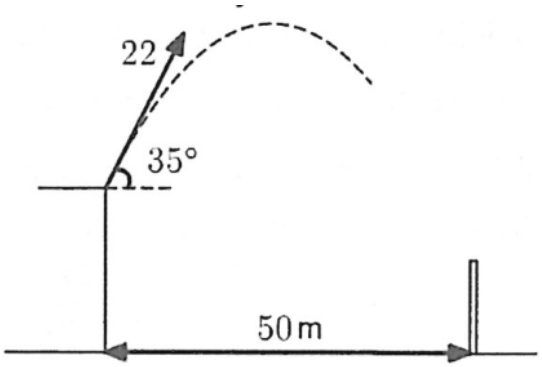
2.58 — Question 5. A stone is thrown from a cliff of height 80m into a lake below. It is projected at 10 ms^{-1} at an angle of α above the horizontal where $\tan\alpha = \frac{4}{3}$



By modelling the stone as a particle, find

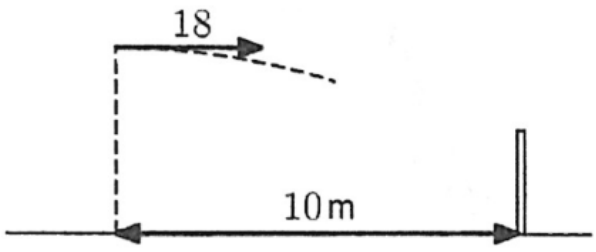
a) the time the stone spends in the air
b) the vertical and horizontal components of the displacement of the stone in terms of t and hence show that the equation of the trajectory of the stone is $y = 80 + \frac{4}{3}x - \frac{49}{360}x^2$
c) the maximum height of the stone above the ground.

2.58 — Question 6. A ball is projected at an angle of 35° above the horizontal from a height of 4.5 metres and at a speed of 22 ms^{-1}. A wall of height 1.5 metres stands 50 metres away.



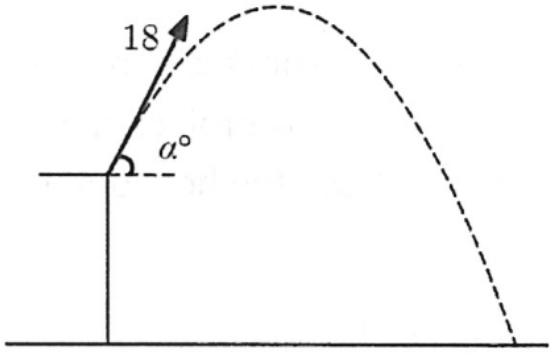
Does the ball hit the wall?

2.58 — Question 7. A tennis ball is hit horizontally with a speed of 18 ms^{-1} from a height of 2.5 metres. The net is 1.2 metres high and is 10 metres away.



a) Will the ball pass over the net?
b) What modelling assumptions have you made?

2.58 — Question 8. A stone is thrown at speed 18 ms^{-1} at an angle of $\alpha°$ where $\tan\alpha = \frac{5}{12}$ above the horizontal from a cliff of height 25 metres.



Modelling the stone as a particle, find the amount of time that the ball is more than 27 metres above ground level.

2.58 — Question 9. A stone is thrown at a speed of u ms^{-1} with angle of α above the horizontal where $\tan\alpha = \frac{3}{4}$ from a cliff of height 40 m. It travels 50 metres horizontally before hitting the water below t seconds later.

a) Find the exact value of t
b) Find the exact value of u
c) State any modelling assumptions you have made.

A stone of twice the mass is thrown from the cliff, otherwise the conditions remain the same.

d) Explain what would change the results in the previous part of the question.

2.58 — Question 10. A particle is thrown at 10 ms^{-1} at an angle of α below the horizontal, where $\tan\alpha = \frac{2}{5}$, from a height of 2 metres. Show that the equation of the trajectory of the particle is

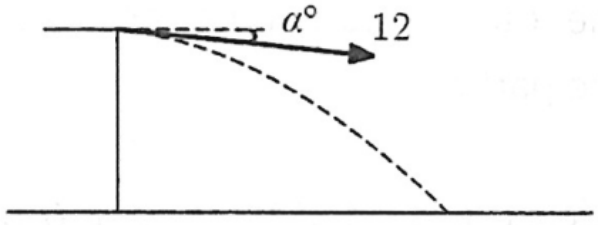
$$y = 2 - \frac{2}{5}x - \frac{1421}{25000}x^2$$

2.58 — Question 11. The following questions use $g = 9.81$ ms^{-2} Dean kicked a football on a horizontal field with speed 20 ms^{-1} at an angle of θ above the horizontal where $\tan\theta = \frac{3}{4}$. The football goal is 36 metres away, the top of the goal post is 2.4 metres above the ground.

a) Does the ball go into the goal?

b) State any modelling assumptions you have made.

2.58 — Question 12. A stone is thrown towards some water at speed 12 ms^{-1} at an angle of $\alpha°$ below the horizontal where $\tan\alpha = \frac{1}{10}$



It lands on the surface of the water 0.4 seconds later. By modelling the stone as a particle, find

a) the direction of the motion when 0.2 seconds have passed.
b) the height above the water surface area that the stone was thrown.
c) the horizontal distance the stone travelled before hitting the water.

2.58 — Question 13. The following questions use $g = 10$ ms^{-2} A ball is projected at an angle of $\theta°$ above the horizontal from a height of 2.5 metres above the ground. The ball hits a target which is 1.5 metres above the ground and 7 metres away from the point of projection, 2 seconds after projection. Modelling the ball as a particle, find

a) the speed at which the ball is projected.
b) the angle at which the ball is projected.

2.58 — Question 14. A ball is projected at angle of 30° above the horizontal from a height of 2 metres at a speed of 18 ms^{-1}. It passes over a wall of height 3 metres stands d metres away. Find

a) the maximum value of d such that the ball still passes over the wall.
b) the direction at which the ball impacts the ground.

2.58 — Question 15. Two balls A and B are projected from a bridge at the same point. Ball A is projected with speed 10 ms^{-1} at an angle of 30° to the horizontal. Ball B is projected 1 second after ball A with speed 40 ms^{-1} at an angle of 60°. By modelling the balls as particles, identify the vertical distance between the balls when one passes over the other, assuming they have not reached the ground.

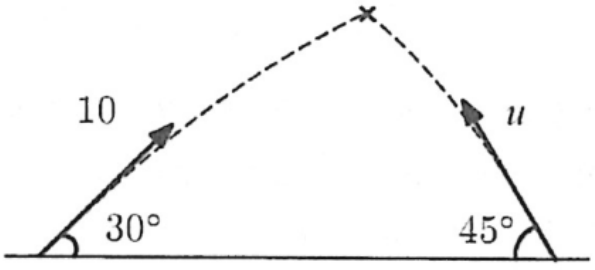
2.58 — Question 16. Prerequisite knowledge for Q16: Reciprocal Trigonometric Functions A child shoots a water pistol, with the water firing at 10 ms^{-1} at an acute angle of $\theta°$ above the horizontal.

a) Show that the equation of the trajectory of the water fired from a height of a metres above the ground is $y = a + x\tan\theta - \frac{x^2}{20\cos^2\theta}$

The water is fired from a height of 0.7 metres above the ground and reaches a horizontal distance of 9.2 metres from the child.

b) Find the angle(s) above the horizontal at which the water pistol was fired.

2.58 — Question 17. The following questions use exact values of g Two balls A and B are projected at the same time towards each other from points P and Q respectively, where P and Q lie 5 metres apart.
Ball A is projected with speed 10 ms^{-1} at 30° to the horizontal.
Ball B is projected with speed u ms^{-1} at 45° to the horizontal.



The two balls collide.

a) Show that the value of u is $5\sqrt{2}$
b) Find the exact time at which they collide giving your answer to 2 significant figures.
c) Find the horizontal distance ball A is from its start point when they collide, giving your answer to 3 significant figures.

2.58 — Question 18. A particle is projected with speed u ms^{-1} from the ground at an angle of θ above the horizontal. Show that the equation of the trajectory of the particle is

$$y = \tan\theta\, x - \frac{g}{2u^2}\sec^2\theta\, x^2$$

2.58 — Question 19. A particle is projected with speed u ms^{-1} at an angle of θ above the horizontal.
The particle reaches a maximum vertical height of h metres.

a) Show that $h = \frac{u^2}{2g}\sin^2\theta$

It is known that $u = 10$ ms^{-1}.

b) Given θ is acute show that $0 \leq h \leq \frac{50}{g}$

Given that a projectile needs to have a maximum height of 5 metres above its start point and is projected at 12 ms^{-1}.
Using $g = 10$ ms^{-2},

c) find the acute angle θ at which it must be projected, giving your answer to 1 decimal place.

2.58 — Question 20. Prerequisite knowledge for Q20: Double Angle Formulae A particle is projected from the ground with speed u ms^{-1} at an angle of θ° above the horizontal.
The particle lands x metres from its original start location after t seconds have passed.

a) Show that $t = \frac{2u}{g}\sin\theta$

b) Show that $x = \frac{u^2}{g}\sin 2\theta$

It is known that the projectile needs to land at least 20 metres away from the start point and is projected at 15 ms^{-1}.
Using $g = 10$ ms^{-2},

c) find the range of possible angles θ at which it must be projected, giving your answers to 1 decimal place.

2.59 — Question 1. A particle's displacement at time t seconds is given by $\mathbf{s}$ metres. In each case find the magnitude of the acceleration of the particle at the given value of t:

a) $\mathbf{s} = (4t^3)\mathbf{i} + (t-3)\mathbf{j}$ at $t = 0.5$

b) $\mathbf{s} = (3t^3 - 5)\mathbf{i} + (5 - t^4)\mathbf{j}$ at $t = 2$

c) $\mathbf{s} = (2e^{0.5t})\mathbf{i} + (t^2)\mathbf{j}$ at $t = 0.1$

2.59 — Question 2. A particle's acceleration at time t seconds is given by $\mathbf{a}$ ms^{-2} and initial velocity $\mathbf{u}$ ms^{-1}. In each case find the speed of the particle at the given value of t:

a) $\mathbf{a} = (4t - 5)\mathbf{i} + (9t^2)\mathbf{j}, \mathbf{u} = 3\mathbf{i} - \mathbf{j}$ at $t = 5$

b) $\mathbf{a} = (12t + 2)\mathbf{i} - 18\mathbf{j}, \mathbf{u} = -2\mathbf{i} + 5\mathbf{j}$ at $t = 0.5$

2.59 — Question 3. A particle's velocity at time t seconds is given by $\mathbf{v}$ metres per second and initial position $\mathbf{s}$ metres. In each case find the distance of the particle from the origin at the given value of t:

a) $\mathbf{v} = (t - 5)\mathbf{i} + 3\mathbf{j}, \mathbf{s} = -4\mathbf{i} + 10\mathbf{j}$ at $t = 0.4$

b) $\mathbf{v} = (3t)\mathbf{i} + (2e^{-0.1t})\mathbf{j}, \mathbf{s} = 2\mathbf{i} + 0.5\mathbf{j}$ at $t = 0.1$

2.59 — Question 4. A particle P of mass 1.2 kg is moving under the action of a single force $\mathbf{F}$ newtons. At time t seconds, the velocity of P, $\mathbf{v}$ ms^{-1} is given by $\mathbf{v} = (5t + 2)\mathbf{i} + (t^2 - 2t)\mathbf{j}$.
When $t = 0$, P is at the point with position vector $(2\mathbf{i} + 3\mathbf{j})$ m. Find

a) the magnitude of $\mathbf{F}$ when $t = 5$
b) the distance from its initial position when $t = 5$

2.59 — Question 5. The velocity of a particle at time t seconds is $\mathbf{v} = (6t^2)\mathbf{i} + (4t - 1)\mathbf{j}$ metres per second. When $t = 1$ it passes a point A and when $t = 5$ it passes the point B. Find the distance AB.

2.59 — Question 6. A particle's position vector at time t seconds is $\mathbf{r} = (2t^2 - t)\mathbf{i} + (2t + 5)\mathbf{j}$ metres.
Find the time at the instant at which the speed of the particle is 2 ms^{-1}.

2.59 — Question 7. The acceleration, ms^{-2}, of a particle at time t seconds is $\mathbf{a} = (2t + 9)\mathbf{i} + (e^{-0.1t})\mathbf{j}$.
When $t = 0$ it passes a point A and when $t = 1$ it passes the point B. Show that the velocity between A and B is $10\mathbf{i} + (10 - 10e^{-0.1})\mathbf{j}$ m.

2.59 — Question 8. A particle's position vector at time t seconds is $\mathbf{r} = (t^2 + t)\mathbf{i} + (5t + 4)\mathbf{j}$ metres.

a) Find the speed of the particle when $t = 4$ seconds.

b) Show that the acceleration of the particle is never zero.

2.59 — Question 9. A particle's velocity at time t seconds is $\mathbf{v} = (t - 4)\mathbf{i} + (t^3 + 2t)\mathbf{j}$ metres per second. The particle is initially at the origin. Find

a) the position vector of the particle in terms of t

b) the vertical displacement when the particle has travelled 10 metres horizontally.

2.59 — Question 10. A particle's velocity at time t seconds is $\mathbf{v} = (2t^{\frac{1}{2}})\mathbf{i} + (3t - 1)\mathbf{j}$ metres per second.
In each case find the values of t at the instant that the particle is moving in the given direction:

a) $\mathbf{i} + \mathbf{j}$
b) $\mathbf{i} - \mathbf{j}$
c) $3\mathbf{i} + 13\mathbf{j}$

2.59 — Question 11. A particle's velocity at time t s is $\mathbf{v} = (2t - 36)\mathbf{i} + (6t^{\frac{1}{2}})\mathbf{j}$ metres per second. It is known that when 1 second has passed, the position vector of the particle is $-20\mathbf{i} + 6\mathbf{j}$. Find

a) the value of t at the instant when the particle is moving in the direction $\mathbf{i} + \mathbf{j}$
b) the position vector $\mathbf{r}$ in terms of t
c) the exact position of the particle at the first instant it is moving with speed 24 ms^{-1}.

2.59 — Question 12. The mass of an object is 0.5 kg. The object is moving with velocity $\mathbf{v} = (6 - pt^{\frac{1}{2}})\mathbf{i} + (5t - q)\mathbf{j}$ metres per second, where t is the number of seconds passed.
Initially, the object has position vector $4\mathbf{i} + \mathbf{j}$
After 4 seconds it passes through $12\mathbf{i} + 21\mathbf{j}$
Find

a) the values of p and q
b) the magnitude of the force acting on the object when $t = 9$ seconds.

2.59 — Question 13. A particle's position $\mathbf{r}$ metres is $\mathbf{r} = (t + \frac{1}{8}k)\mathbf{i} + (kt^2 - 8t)\mathbf{j}$ where t is the time in seconds. At the instant that $t = 0.5$, the particle is moving parallel to $\mathbf{i}$. Find

a) the value of k
b) the value of t when the particle is 1 metre from the origin.

2.59 — Question 14. The position vectors of particles P and Q are

$$\mathbf{r}_P = (3 - t^2)\mathbf{i} + (3t^2 + 4)\mathbf{j} \qquad \mathbf{r}_Q = (12 - 2t^2)\mathbf{i} + (4t^2 + 5)\mathbf{j}$$

respectively, where t is time in seconds.

a) Show that the distance between P and Q at any given time satisfies $d^2 = 2t^4 - 16t^2 + 82$

b) Find the time at which the distance between P and Q is minimised during the motion.

2.59 — Question 15. The position vectors of particles P and Q are $\mathbf{r}_P = (t + 1)\mathbf{i} + (6t)\mathbf{j}$, $\mathbf{r}_Q = (4t^2)\mathbf{i} + (5t + 1)\mathbf{j}$ respectively, where t is the time in seconds.

Find the exact value of t when both particles are moving at the same speed.

2.59 — Question 16. The velocity vectors of particles P and Q are $\mathbf{v}_P = (4t)\mathbf{i} + (t^2 + \frac{2}{3})\mathbf{j}$, $\mathbf{v}_Q = (2t + 7)\mathbf{i} + (-4)\mathbf{j}$ respectively, where t is the time in seconds.
They are both initially at the origin.
Show that particle Q is four times particle P's distance from the origin when $t = 1$

2.59 — Question 17. The following questions use that i and j are unit vectors pointing east and north respectively. A particle's position $\mathbf{r}$ metres is $\mathbf{r} = (5e^{0.4t} + 1)\mathbf{i} + (t^2 - 4t)\mathbf{j}$ where t is the time in seconds.

a) Find an expression for the velocity of the particle at time t
b) State the direction in which the particle is travelling when $t = 2$
c) The mass of the particle is 10 kg. Find the magnitude of the resultant force on the particle when $t = 2$

2.59 — Question 18. The mass of an object is 2 kg. The object is moving with velocity $\mathbf{v} = (t^2 + pt)\mathbf{i} + (4t^3)\mathbf{j}$ metres per second, where t is the number of seconds passed since passing through the origin. The magnitude of the force acting on the object is 120 N when $t = 2$ seconds.

a) Show that p satisfies $p^2 + 8p - 1280 = 0$

b) Hence, find the possible value of p

c) Show that there is no instantaneous time for which the object moves eastwards or westwards.

2.59 — Question 19. A particle's velocity at time t seconds is $\mathbf{v} = (2t^2)\mathbf{i} + (-5t)\mathbf{j}$ metres per second.
Find the bearing the particle is travelling on when $t = 0.1$ seconds to the nearest degree.

2.59 — Question 20. An object's velocity $\mathbf{v}$ metres per second is $\mathbf{v} = (-t^2 - 4)\mathbf{i} + (-t^3)\mathbf{j}$ where t is the time in seconds.
Find the time(s) at which the object is moving at a bearing of 225°

2.59 — Question 21. An object's position $\mathbf{r}$ metres is given by $\mathbf{r} = (t + \frac{7}{4})\mathbf{i} + (2 - t^2)\mathbf{j}$ where t is the time in seconds. Find

a) the time(s) at which the object is at a bearing of 135° to the origin.
b) the speed of the object when it is directly east of the origin.

2.59 — Question 22. A particle's position $\mathbf{r}$ metres is $\mathbf{r} = (3 + t^2)\mathbf{i} + (8t - 5)\mathbf{j}$ metres where t is the time passed in seconds.
At time T, in the instant the object is moving with a bearing 045°.
Find

a) an expression for the velocity in terms of t
b) the value of T and the speed of the object at your value of T

2.59 — Question 23. An object of mass 0.5 kg is initially at rest. It is being pushed by forces $\mathbf{F}_1 = -3\mathbf{i}+(t^2-3t+7)\mathbf{j}$ N and $\mathbf{F}_2 = (t^2 - 1)\mathbf{i} - 5\mathbf{j}$ N where t is the number of seconds passed.
Find

a) the value of t when the object is in equilibrium.
b) the instantaneous time at which the object is moving due north-east.
c) the speed of the object when $t = 1$

2.59 — Question 24. The following questions use Calculus from non-AS-Level content A particle's displacement at time t seconds is $\mathbf{s} = (2^{2t})\mathbf{i} - (2^t)\mathbf{j}$ metres. Find

a) the speed of the particle when $t = 0.4$

b) the time at which the particle is at a bearing of $135°$ to the origin.

2.59 — Question 25. A particle's acceleration in ms^{-2} at time t seconds is $\mathbf{a} = 8\mathbf{i} - (\frac{2}{t})\mathbf{j}$.
When $t = 1$, the particle is moving with velocity $(3\mathbf{i} - 5\mathbf{j})$ ms^{-1}. Find

a) an expression for the velocity in terms of t

b) the speed of the particle at $t = 0.1$

2.59 — Question 26. A particle's velocity at time t seconds is $\mathbf{v} = (e^t)\mathbf{i} + (\sin t)\mathbf{j}$ metres per second. When $t = 0$, the particle is $(2\mathbf{i} + 5\mathbf{j})$ m from the origin.
Find the distance of the particle from the origin when $t = 0.5$

2.59 — Question 27. A particle's position in relation to the origin is $\mathbf{r} = (4\cos(2t))\mathbf{i} + (t^2 \sin t)\mathbf{j}$ metres where t is the time in seconds. The mass of the particle is 0.5 kg.

a) State the position of the particle in relation to the origin when $t = 3\pi$
b) Find the direction in which the particle is travelling when $t = 3\pi$
c) Find the magnitude of the resultant force on the particle when $t = 3\pi$

2.59 — Question 28. A particle's acceleration at time t is $\mathbf{a} = 6\mathbf{i} + (te^{2t})\mathbf{j}$ metres per second squared.
The initial velocity of the particle is $(\frac{3}{4}\mathbf{j})$ ms^{-1}.
Find the speed of the particle when $t = 0.2$ seconds.

2.59 — Question 29. The velocity of a particle at time t seconds is $\mathbf{v} = (\sin 2t)\mathbf{i} + (\cos t)\mathbf{j}$ metres per second.
When $t = 0$ it passes a point A and when $t = \frac{\pi}{2}$ it passes the point B.
Find the distance AB.

2.59 — Question 30. A particle's position at time t seconds is $\mathbf{r} = (\cos t - e^t)\mathbf{i} + (\sin t - \ln t)\mathbf{j}$ metres.
Find the bearing the particle is travelling on when $t = 0.5$ seconds to the nearest degree.

2.59 — Question 31. Prerequisite knowledge for Q31–Q32: Parametric Equations For each of the following, write a cartesian equation for the path of the motion:

a) $\mathbf{r} = (8t - 5)\mathbf{i} + (3 + t^2)\mathbf{j}$
b) $\mathbf{r} = (t + \frac{7}{4})\mathbf{i} + (2 - t^2)\mathbf{j}$
c) $\mathbf{r} = (t^2)\mathbf{i} + (5t + 4)\mathbf{j}$

2.59 — Question 32. A particle P has mass 2 kg and its position is given by $\mathbf{r}$ metres $\mathbf{r} = (\sin t)\mathbf{i} + (\cos t)\mathbf{j}$ where t is the time in seconds.

a) Find the magnitude of the resultant force on the particle when $t = \frac{\pi}{2}$

b) Write a cartesian equation for the path of the motion.